Lucky Jain

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Education

B.S. Computer Science & Neuroscience | *Johns Hopkins University* Aug 2024 – May 2027
Dean's List
Relevant Courses: Neural Signals & Computation, Probability & Statistics, Linear Algebra & Differential Equations

Experiences

Researcher | *Cognitive Neurophysiology and BMI Lab* Oct 2025 – Present
- Built an end-to-end signal-processing and deep-learning pipeline for continuous cursor decoding from 1,024-channel human ECoG.
- Developed computer-vision and geometry pipeline to localize high-density cortical electrode arrays from 2D intraoperative images onto 3D brain surface meshes.

Researcher | *Radiation Oncology at Johns Hopkins Medical Institute* Oct 2025 – Present
- Built a human-in-the-loop automation workflow to accelerate Epic chart abstraction.
- Curated longitudinal Epic data for 500+ spine stereotactic body radiation therapy (SBRT) patients at the Johns Hopkins Hospital.
- Observed radiation oncology consultations, treatment design, and cross-team coordination.

Decoding Lead | *Johns Hopkins Brain-Computer Interface Society* Sep 2024 – June 2026
- Designed EEG decoding pipeline including multi-band preprocessing, CSP+LDA and power-based models, cross-validation, temporal window sweeps, and fatigue analyses.
- Led data collection on 15+ subjects for novel low-resolution motor imagery experiment.

Researcher | *Laboratory for Computational Intensive Care Medicine* Nov 2024 – June 2026
- Engineered and validated 10+ clinically motivated time-series features (e.g. infections, medications, ventilation) for dynamic machine learning model predicting ICU delirium.
- Extracted, cleaned, and structured ICU electronic health record data from Mayo Clinic (50,000+ patients) under computation and memory limits.

Co-Creator | *Eyedentify-AI* May 2025 – Feb 2026
- Built a deep-learning computer vision diagnostic system (94% accuracy) with HuggingFace Transformers and Grad-CAM explainability; optimized backend for real-time inference; deployed full-stack platform to live users.
- Presented at BMES and DSAI conferences and supported by the JHU Spark accelerator.

Skills

Signal processing | Machine learning | Neural data analysis
Python (PyTorch, NumPy, Pandas, OpenCV)